Swapnamoy Garai

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Professional Summary

Software Development Engineer pursuing B.Tech in CSE at Jadavpur University, interning at Quantiphi Analytics. Proven track record architecting **cloud-native microservices**, **clinical healthcare ML pipelines**, and **Agentic AI/NLP platforms**. Experienced in deploying production ML models (XGBoost, SHAP, BERT/LaBSE), developing high-concurrency APIs in **Python** and **Go**, and utilizing GenAI developer tools (Copilot, Claude) to accelerate enterprise software delivery.

Technical Skills

Languages: **Python**, **SQL**, **Go (Golang)**, **C++**, **TypeScript/JavaScript**, C#, Rust, C
AI, NLP & GenAI: **Agentic AI**, **NLP**, **LLM Orchestration**, Model Context Protocol (**MCP**), **BERT**, **LaBSE**, Prompt Engineering, Copilot, Claude AI
Cloud, Systems & DevOps: **Docker**, **Kubernetes (K8s)**, **Azure / Cloud Services**, **FastAPI**, **SQL Server / PostgreSQL**, **Redis**, CI/CD Automation
ML & Healthcare Analytics: **XGBoost**, **SHAP (Explainable AI)**, **MIMIC-III Clinical Data**, PyTorch, Scikit-Learn, Pandas
Software Practices: **Test-Driven Development (TDD)**, Microservices, Agile, REST APIs, gRPC, Git, Linux/WSL

Work Experience

Quantiphi Analytics Bengaluru, India
Software Developer Intern *Jun 2026 – Present*

- Engineered production **FastAPI REST microservices** following OpenAPI specifications, integrating with **PostgreSQL** and **Redis** for reliable cloud-native data processing.
- Implemented **Test-Driven Development (TDD)** and automated **CI/CD pipelines**, enforcing code quality gates to maintain an **85%+** test suite pass rate.
- Designed dimensional star schemas and optimized transformation SQL with window functions for enterprise data pipelines.
- Leveraged GenAI-assisted tooling (Claude, Copilot) to accelerate feature prototyping and automate integration testing workflows.

Crispat Remote
Full Stack Developer Intern *Jun 2024 – Sep 2024*

- Built full-stack web applications with **React.js**, **Node.js**, and **Express**, utilizing **Redis** for real-time state synchronization.
- Architected a scalable multipart file-handling service, slashing document upload latency by **50%** through concurrent streaming.
- Implemented interactive connection graph visualizations with **React Flow** and **D3.js**, enhancing user engagement by **40%**.

Key AI & Software Engineering Projects

Clinical Length of Stay (LOS) AI Platform | GitHub | *Healthcare AI Engine* Python, XGBoost, SHAP, FastAPI, Next.js

- Developed an end-to-end **clinical AI prediction platform** forecasting patient hospital Length of Stay (LOS) across **10,000+ patient records** from the MIMIC-III database.
- Engineered an **XGBoost ML pipeline** coupled with **SHAP explainability**, providing clinicians transparent feature attribution for critical lab markers (Glucose, WBC, platelets).
- Built cloud-ready **FastAPI** microservices serving real-time predictions and patient-flow telemetry to a reactive Next.js dashboard evaluated on a 20% held-out test split.

Multilingual NLP Detection Platform | GitHub | *Deep Learning & NLP Pipeline* Python, PyTorch, BERT, LaBSE

- Engineered a cross-lingual NLP pipeline utilizing a **hybrid CNN-LSTM + Attention** model across English, Bengali, Japanese, and Arabic text streams.
- Solved the long-document context window constraint by building a **Sliding Window Segmenter** (1,024 tokens), processing text 4x longer than standard transformer limits.
- Leveraged **LaBSE sentence embeddings** to maintain high cross-lingual semantic fidelity, achieving **93.31% accuracy** and outperforming the baseline by 18%.

Py-Sandbox-Agent | GitHub | *Autonomous Agentic AI & Sandboxed Execution* Python, Docker, AST, FastAPI

- Architected an autonomous **Agentic AI system** featuring an FSM self-correction loop that parses execution tracebacks and iteratively refines generated code against unit test assertions.
- Enforced strict runtime safety using an **AST security scanner** (default-deny imports, blocking eval/exec) and **Docker container isolation** (network-gapped, 256MB memory cap, ulimit).
- Implemented a **100-slot concurrency pool** with FIFO backpressure queueing, preventing resource exhaustion during peak agent workloads.

Education

Jadavpur University Kolkata, India
B.Tech in Computer Science & Engineering ***(CGPA: 8.4 / 10.0)*** *Nov 2022 – June 2026*

Honors & Certifications

Competitive Programming: Solved **800+ LeetCode problems** (Peak Rating: **2000**, Best Contest Rank: **208**). Active on Codeforces & AtCoder.
Standardized Exams: **JEE Advanced AIR 5576** | **WBJEE Rank 207** (Top 0.2% among 100,000+ candidates).
Certification: **Claude Certified Developer** – Foundations Certified.